Write a program in Java for sorting a given list of strings in ascending order.

CODE:

```
import java.util.Scanner;

import java.util.Arrays;


public class Pract10aSortStrings {

   public static void main(String[] args) {


      Scanner sc = new Scanner(System.in);


      System.out.print("Enter number of strings: ");

      int n = sc.nextInt();

      sc.nextLine();

      String[] arr = new String[n];

      System.out.println("Enter the strings:");

      for (int i = 0; i < n; i++) {

         arr[i] = sc.nextLine();

      }

      Arrays.sort(arr);

      System.out.println("Strings in Ascending Order:");

      for (String s : arr) {

         System.out.println(s);

      }

      sc.close();

   }

}
```

Write a program in Java to demonstrate hierarchical inheritance.

CODE:

```
class Animal {
   void eat() {
      System.out.println("Animal eats food");
   }
}
class Dog extends Animal {
   void bark() {
      System.out.println("Dog barks");
   }
}
class Cat extends Animal {
   void meow() {
      System.out.println("Cat meows");
   }
}
public class HierarchicalInheritance{
   public static void main(String[] args) {
      Dog d = new Dog();
      d.eat(); // inherited method
      d.bark(); // Dog's own method

      Cat c = new Cat();
      c.eat(); // inherited method
      c.meow(); // Cat's own method
   }
}
```

Write a program in Java to find the biggest element among three numbers using the ternary operator.

CODE:

```
import java.util.Scanner;


public class TernaryOp{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.print("Enter third number: ");
        int c = sc.nextInt();


        int biggest = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);


        System.out.println("Largest number: " + biggest);


        sc.close();
    }
}
```

Write a program in Java to implement multiple inheritance using the interface.

CODE:

```
interface A {
   void showA();
}
interface B {
   void showB();
}
class C implements A, B {

   public void showA() {
      System.out.println("Method of Interface A");
   }
   public void showB() {
      System.out.println("Method of Interface B");
   }
   void showC() {
      System.out.println("Method of Class C");
   }
}
public class MultipleInheritance {
   public static void main(String[] args) {
      C obj = new C();
      obj.showA();
      obj.showB();
      obj.showC();
   }
}
```

Write a program in Java to find the area and perimeter of a circle using an abstract class.

CODE:

```
abstract class Shape {
   abstract void area();
   abstract void perimeter();
}
class Circle extends Shape {
   double radius;
   final double PI = 3.14159;

   Circle(double r) {
      radius = r;
   }
   void area() {
      System.out.println("Area of Circle = " + (PI * radius * radius));
   }
   void perimeter() {
      System.out.println("Perimeter of Circle = " + (2 * PI * radius));
   }
}
public class AbstractCircleDemo {
   public static void main(String[] args) {
      Circle c = new Circle(5);
      c.area();
      c.perimeter();
   }
}
```

Write a program in Java to find the average of three numbers using method overriding.

CODE:

```
class Average {
    void findAverage(int a, int b, int c) {
        System.out.println("Average method in parent class");
    }
}
class CalculateAverage extends Average {
    @Override
    void findAverage(int a, int b, int c) {
        double avg = (a + b + c) / 3.0;
        System.out.println("Average of three numbers: " + avg);
    }
}
public class Pract6cMethodOverridingDemo {
    public static void main(String[] args) {
        Average obj = new CalculateAverage();
        obj.findAverage(10, 20, 30);
    }
}
```

Write a program in Java to find the sum and average of three numbers using the super keyword.

```
class Numbers {
   int a, b, c;
   Numbers(int a, int b, int c) {
      this.a = a;
      this.b = b;
      this.c = c;
   }
   int sum() {
      return a + b + c;
   }
}
class Calculate extends Numbers {
   Calculate(int a, int b, int c) {
      super(a, b, c);
   }

   void display() {
      int total = super.sum();
      double average = total / 3.0;
      System.out.println("Sum of three numbers: " + total);
      System.out.println("Average of three numbers: " + average);
   }
}

public class Super{
   public static void main(String[] args) {
```

```
        Calculate obj = new Calculate(10, 20, 30);
        obj.display();
    }
}
```

Write a program in Java to check the grade of marks using a switch case.

```
import java.io.BufferedReader;
import java.io.IOException;
import java.io.InputStreamReader;

public class GradeCheck{
    public static void main(String[] args) throws IOException {
        System.out.print("Enter marks (0-100):");
        BufferedReader reader = new BufferedReader(new InputStreamReader(System.in));
        String inputString = reader.readLine();
        int marks = Integer.parseInt(inputString);

        switch (marks / 10) {
            case 10:
            case 9:
                System.out.println("Grade: A");
                break;
            case 8:
                System.out.println("Grade: B");
                break;
            case 7:
                System.out.println("Grade: C");
                break;
            case 6:
```

```
            System.out.println("Grade: D");
            break;
        default:
            System.out.println("Grade: F");
            break;
    }
  }
}
```

Write a program in Java to multiply two matrices.

```
import java.util.Scanner;
public class MatrixMultiplication {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter rows of first matrix: ");
    int r1 = sc.nextInt();
    System.out.print("Enter columns of first matrix: ");
    int c1 = sc.nextInt();

    System.out.print("Enter rows of second matrix: ");
    int r2 = sc.nextInt();
    System.out.print("Enter columns of second matrix: ");
    int c2 = sc.nextInt();

    if (c1 != r2) {
      System.out.println("Matrix multiplication not possible (c1 != r2).");
      sc.close();
       return;
```

```
}
int[][] A = new int[r1][c1];
int[][] B = new int[r2][c2];
int[][] C = new int[r1][c2];

System.out.println("Enter elements of Matrix A:");
for (int i = 0; i < r1; i++) {
    for (int j = 0; j < c1; j++) {
        A[i][j] = sc.nextInt();
    }
}
System.out.println("Enter elements of Matrix B:");
for (int i = 0; i < r2; i++) {
    for (int j = 0; j < c2; j++) {
        B[i][j] = sc.nextInt();
    }
}
for (int i = 0; i < r1; i++) {
    for (int j = 0; j < c2; j++) {
        for (int k = 0; k < c1; k++) {
            C[i][j] += A[i][k] * B[k][j];
        }
    }
}
System.out.println("Resultant Matrix C (A × B):");
for (int i = 0; i < r1; i++) {
    for (int j = 0; j < c2; j++) {
        System.out.print(C[i][j] + " ");
```

```
            }
            System.out.println();
        }
        sc.close();
    }


}
```

Write a Java program to implement a thread.